Practical 2 – Problem sheet

Exercise 1

Write a program that evaluates the following functions of a real variable x. Since these functions are piecewise defined, you will need conditional statements in your program. See the code from Lesson 5, class `Conditionals`, for examples.

S $$f_1(x) = \begin{cases} 3x+4 & \text{if } x \geq 2, \\ 0 & \text{otherwise;} \end{cases}$$

S $$f_2(x) = \begin{cases} \frac{x}{3} & \text{if } x \geq 1, \\ \frac{x}{5} & \text{otherwise;} \end{cases}$$

S $$f_3(x) = \begin{cases} x^2 & \text{if } x \geq 5, \\ 2x^3 & \text{if } 1 \leq x < 5, \\ 3x^4 & \text{if } x < 1; \end{cases}$$

I $$f_4(x) = \begin{cases} e^{x/2} & \text{if } x \in (-1,3), \\ e^{-2x} & \text{otherwise;} \end{cases}$$

I $$f_5(x) = \begin{cases} 2\cos x & \text{if } \cos x > \frac{1}{2}, \\ -\sin 2x & \text{otherwise;} \end{cases}$$

The following functions are not necessarily defined for all $x \in \mathbb{R}$. If x is not in the domain of the function, your program should return `Double.NaN` as the result.

I $$f_6(x) = \frac{\sin x}{x} \qquad \text{for } x > 0;$$

I $$f_7(x) = 4^{x+4} \arctan x \qquad \text{for } x \in [-3,5];$$

A $$f_8(x) = \cosh^{-1}\left(\frac{y^2}{y^2-1}\right) \qquad \text{with } y = x^4 - 4x + 1,$$

defined in the region where the denominator is positive.

Exercise 2 S

Consider the quadratic equation $ax^2 + bx + c = 0$ with given real coefficients a, b, c. It is well known that the two real solutions of the equation are

$$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}, \quad \text{where } \Delta = b^2 - 4ac,$$

provided that the "discriminant" Δ is positive. If Δ is negative, then there is no real solution, and we will ignore the case $\Delta = 0$ here (this case would be hard to detect in a computer program due to possible roundoff errors).

Write a program that, given real numbers a, b, c (in floating point format), decides whether real solutions of the equation $ax^2 + bx + c = 0$ exist, and if so, computes these two solutions. The program should output a character string as follows:

- If no real solution exists: `"no real solutions"`
- If real solutions exist:
  `"real solutions:  -1.234567890, 1.234567890"`
  (The numbers are examples; the program should insert the actual solutions. The smaller solution should go first.)

A code template is provided for you, containing already the rough outline of the program. Complete it by adding code at the places marked "`...`". You may also want to review the example code from Lesson 1, class `Roots`.

Exercise 3 (I)

A few years ago, the Higher Education Academy proposed to replace the system of degree classes at UK universities (1st class, upper 2nd class, etc.) by a new "Grade Point Average" system (GPA); see here. Under the proposed system, degrees would be classified in grades A+, A, A–, B+, . . . , F; see the "guide for students" available on Moodle. Your task is to convert a grade from the University of York mark scale (0–100) to these new GPA grades.

Write a program that takes a mark on the University of York mark scale (0–100) as an input, and produces the corresponding GPA grade (e.g. B+) as an output. The input of the program should be a floating point number, the output should be a character string. Marks are converted by the following rules:

- Integer marks on the 0–100 scale are mapped to new grades as described in the "guide for students" (see Moodle).
- Fractional marks are understood to be rounded to the nearest integer.
- If the input value is outside the UoY mark scale, return `"undefined"` as the output.

For guidance, you can refer to the example code from Lesson 5, class `Conditionals`, function `degreeClass`.

Exercise 4 (I)

Revisit Exercise 4 from the Practical 1 exercise sheet. Modify the function `timeConversion()` so that it accounts for plurals correctly: instead of "1 hours", it should output "1 hour", and correspondingly for minutes and seconds. All other functionality should stay the same. For example, for input value `3602`, the output should be:

```
It is 1 hour, 0 minutes and 2 seconds since midnight.
```

Start from the code template `Time` (this is essentially a copy of the model solutions for Practical 1, Exercise 4).

Exercise 5 (S)

In this exercise, you will write small programs that use loops in order to compute certain mathematical expressions. Code templates are given; some of these suggest to use `while` loops and others suggest `for` loops, but you can use either type of loop for this exercise. For examples, see the sample code from Lesson 7, class `Loops`. Use floating point arithmetic for all computations.

(a) Compute $\sum_{n=1}^{20}(n^3 - 2)$.

(b) Compute $\sum_{j=2}^{100}\log(j^3 - 2)$.

(c) Compute $\sum_{j=1}^{n}\sin(4j)$, where the integer $n \geq 1$ is given as a parameter.

In each case, the return value of the corresponding function should be the value of the sum in question.

Exercise 6 (I)

By a famous result of Euler, one has

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}. \tag{1}$$

Let us verify this formula numerically. Write a program that computes, for given $N \in \mathbb{N}$, the sum

$$S(N) := \frac{6}{\pi^2}\sum_{k=1}^{N}\frac{1}{k^2}. \tag{2}$$

Check whether $S(N)$ is near to 1 for large N. A code template is provided (`EulerSum`).

Hint: If you're running into problems at very large N, check carefully where integer arithmetic and where floating point arithmetic are applied, and which effects this may have.

Exercise 7 (A)

Essentially all products in today's supermarkets are labelled with barcodes. These encode the European Article Number (EAN-13) of the product, a 13-digit unique identifier number.

(a) Read about the EAN-13 system, in particular, on how the barcodes are constructed and about the usage of a check digit. You can find some information here.

(b) Write a function `isValidEAN()` that checks whether a number (given as long integer) is a valid EAN code. It should return a boolean value.

(c) Write a function `constructBarcode()` that computes the barcode for a given EAN. Here the barcode should be formatted as a string of 95 characters that are either `X` (for black) or `_` (for white).